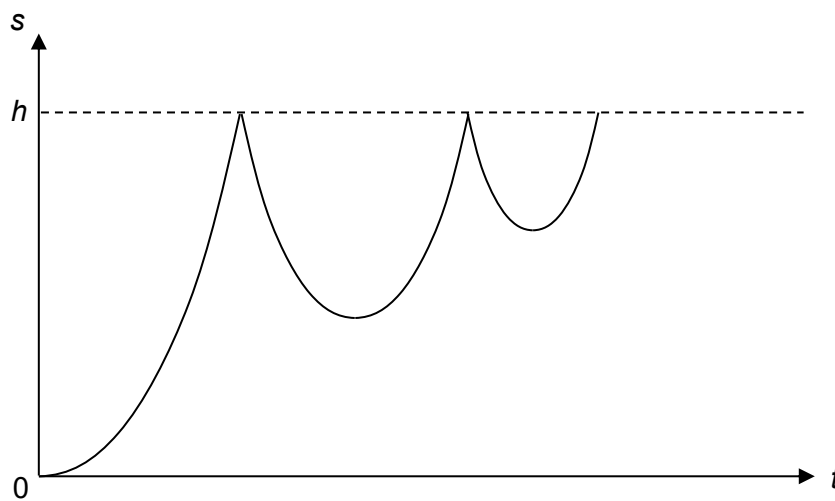
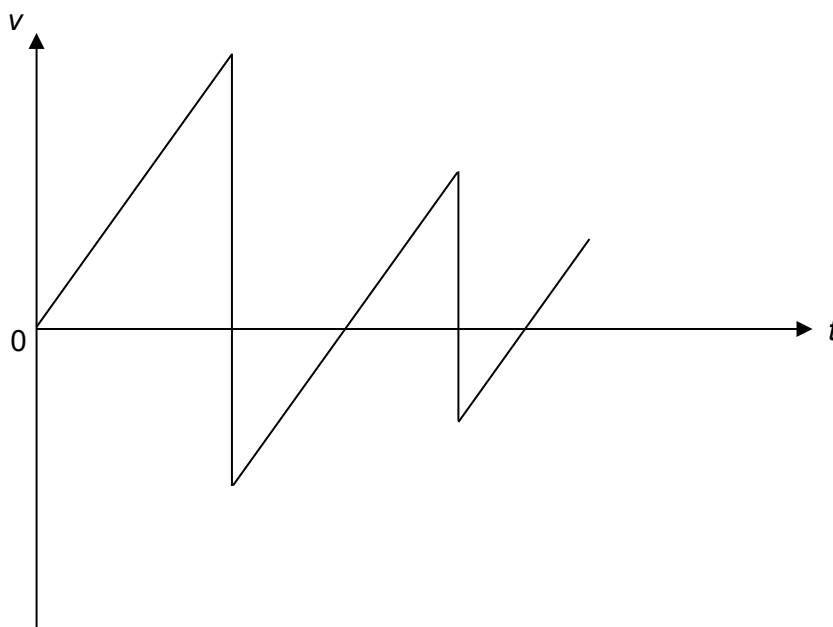


2 (a) (i)



Shape (including the duration interval is decreasing) and start from 0 [2]

(ii)



Shape [1]
Decreasing peak [1]

(b) (i) $mgh = \frac{1}{2}mv^2$
 $h = \frac{(1/2)(3.8^2)}{(9.81)} = 0.736 \text{ m}$ [1]

(ii) Using Newton's 2nd Law, average *net* force = $\Delta p / \Delta t$
 $= (0.062)(1.7 + 3.8) / 0.081$
 $= 4.2 \text{ N}$ [1]

Average force ground acts on ball = $4.2 + mg$
 $= 4.8 \text{ N}$ [1]

(c) Horizontally, $(3.0 \cos 45^\circ)(t) = 0.20$
 $t = (0.20) / (3.0 \cos 45^\circ)$ [1]

Vertically, $(3.0 \sin 45^\circ) = (-3.0 \sin 45^\circ) + (a)[(0.20) / (3.0 \cos 45^\circ)]$
 $a = 45 \text{ m s}^{-2}$ [1]

$$q(V / d) - mg = ma$$

$$(4.4 \times 10^{-7})(V / 0.10) - (1.0 \times 10^{-4})(9.81) = (1.0 \times 10^{-4})(45)$$

Solving, $V = -1250 \text{ V}$ [1]